

AMIYA KUMAR MISHRA

AI Engineer — Agentic AI, LLMs & RAG Systems

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Professional Summary

AI Engineer with 4 years of experience designing and deploying **agentic AI systems**, multi-agent architectures, and production-grade RAG platforms. Hands-on expertise with **Python, LangChain, LangGraph-style workflows, tool calling, function calling, and autonomous agent coordination**. Built scalable hybrid-RAG pipelines using FAISS, SGI-based retrieval, and transformer-powered semantic chunking. Experienced in architecting backend systems, building AI developer tools, and deploying real-world AI assistants across enterprise environments.

Technical Skills

Agentic AI: Multi-Agent Architectures, Tool Calling, Function Calling, Agent Graphs, Autonomous Agents, LangGraph (familiar), LangChain, Agent Workflow Orchestration.

RAG & Retrieval: Hybrid RAG Pipelines, FAISS, Chroma, SGI APIs, Semantic Chunking (Transformer-based), Query Routing, Context Grounding.

LLM Ecosystem: OpenAI, Mistral, Prompt Engineering, Structured Outputs, Context Windows, Token Optimization.

Backend Development: Python, FastAPI (familiar), Flask, REST APIs, WebSockets, JWT Authentication, Async Pipelines.

Frontend/UI: Vue.js 2/3.

DevTools Engineering: VSCode Extensions, Model Context Protocol (MCP), CLI Tools, Developer Automation.

Cloud/DevOps: AWS EC2, Docker, Kubernetes, GitLab CI/CD, Nginx.

Data Handling: MongoDB, SGI Retrieval APIs, FAISS Vector DB, NumPy, Pandas, Regex, XML/JSON Parsing.

Professional Experience

Dassault Systèmes Global Services — Bangalore

Software Engineering Specialist

May 2024 – Present

- Transformed the Mining Assistant into a **fully agentic AI system** with multiple autonomous agents employing tool calling and function calling for structured reasoning and action.
- Designed a **LangGraph-style multi-agent orchestration framework** where planner, retriever, verifier, summarizer, and executor agents communicate via structured message passing and state transitions.
- Integrated **SGI database APIs** enabling agents to perform real-time semantic graph traversal and dynamic DB/API lookups for complex mining insights and evidence-backed responses.
- Implemented **hybrid RAG pipelines** combining SGI graph retrieval with FAISS vector search to maximize contextual relevance and reduce hallucination risk.
- Built tool interfaces for agent capabilities: data extraction, query execution, code/tool invocation, verification, and summarization.
- Developed scalable backend services in Python/Flask with asynchronous inference pipelines and JWT-secured access control.
- Built a real-time interactive dashboard in Vue.js to visualize agent reasoning steps, tool calls, and retrieval provenance.

- Led containerized deployment using Docker, Kubernetes, and AWS EC2 to ensure production reliability and observability.
- **Award: Genius Award — H2 2024** for advancements in agentic AI automation.

Software Engineering Associate

Jul 2022 – Apr 2024

- Developed an AI-powered Aviation Chatbot Platform for aerospace clients.
- Built a **FAISS-powered predictive analytics retrieval engine**, optimizing semantic relevance and query latency for aviation datasets.
- Implemented REST APIs in Flask and integrated structured datasets for safe retrieval and inference.
- Developed secure Vue.js components for domain-specific UI and role-based access to AI features.

Software Engineering Intern

Jan 2022 – May 2022

- Contributed to UI components and API prototyping for internal Data and AI R&D tools, gaining foundational software engineering experience.

Key Projects

Agentic Hybrid RAG Mining Assistant

- Upgraded the mining assistant into a **multi-agent architecture** supporting tool calling, function calling, and coordinated reasoning across autonomous agents.
- Designed an **agent graph pipeline** (planner, retriever, verifier, summarizer, executor) with LangGraph-style orchestration and state transitions.
- Integrated SGI database APIs enabling agents to perform **real-time semantic graph traversal + dynamic data lookup**, improving factual grounding.
- Implemented hybrid retrieval (SGI graph traversal + FAISS vector search) for robust domain-specific responses.
- Built real-time transparency features in the UI to display agent call traces, retrieval sources, and verification steps.

Multi-Agent DevTool Assistant (VS Code MCP Extension)

Hackathon Winner

- Developed a fully autonomous **multi-agent developer assistant** for code review, refactoring, documentation generation, and test creation.
- Built a VSCode extension using **Model Context Protocol (MCP)** with modular tool and context providers.
- Implemented agentic orchestration using Python/Flask backend integrated with OpenAI and Mistral models.
- Winner of Ideathon + Hackathon 2025; approved for internal productization.

Aerospace Production Line Assistant

- Designed and deployed a production-grade Aerospace Production Line Assistant integrating with enterprise APIs and centralized knowledge systems.
- Enabled production engineers, staff, and managers to access operational data, workflows, and process insights through a unified AI-driven platform.
- Developed intelligent retrieval and workflow support capabilities to streamline production-line decision making and information accessibility.

Intelligent Cloud Configurator

- Developed a multi-agent system for infrastructure teams to analyze, manage, and optimize cloud configurations across enterprise environments.
- Implemented autonomous tenant provisioning workflows capable of configuring new cloud tenants based on user and infrastructure requirements.
- Designed agent orchestration and decision-making pipelines to streamline cloud setup, validation, and configuration management processes.

Education

B.Tech – Computer Science & Engineering

Centurion University of Technology and Management, Bhubaneswar

2018–2022

CGPA: 9.0

Certifications & Awards

- Building Personalized Chatbots with OpenAI & LangChain — DeepLearning.AI (2024)
- Genius Award — Dassault Systèmes (2024)
- Winner — Ideathon + Hackathon 2025 (Multi-Agent DevTool Assistant)
- Introduction to AI — LinkedIn Learning